How to work

- Use the same `stata/do/` folder as in weeks 1 and 2: the new worksheets are in `exercises/`. In each worksheet, write the path of your own `stata/do` folder in the `cd` line.
- Each worksheet has two parts. Part A contains the Guided Stata commands shown on the slides, in the same order. Part B is the laboratory: save the worksheet under a new name, complete the TODO lines and write your answers as comments.
- **Predict before you run**: for every PREDICT slide, write your guess before executing the command.
- Every coefficient is written as a sentence: its unit, the comparison group, and what is held fixed. Every number comes with the number of observations behind it.
- Laboratory work is formative and not graded.

Data for week 3

The files of weeks 1 and 2: `data/justice_europe_2019.dta` and `data/contracts_2019.dta`.

Lab 5: multiple regression

Questions. Do countries with more police record less theft once we compare countries with similar income? What does a coefficient on an income-group indicator measure?

Worksheet. `exercises/05_multiple_regression.do`.

Before class: reading

Berk, J. B., Harvey, C. R. and Hirshleifer, D. (2017), How to write an effective referee report and improve the scientific review process, *Journal of Economic Perspectives* 31(1): 231–244, doi:10.1257/jep.31.1.231. The whole article (short). Bring notes on:

1. Which three problems of the review process do the authors identify?
2. Why do they call the failure to separate essential comments from suggestions the highest-cost mistake?
3. What is the question "Flaws and all, would I have been pleased to have written such a paper?" for?

Hand in your one-page outline (question, data and method of the paper you are reviewing) at the end of the lecture.

Tasks

1. Regress recorded theft on police, then add log GDP per capita. Write each police coefficient as a sentence with units; in the second, say what is held fixed.
2. Check the omitted variable formula: simple coefficient = multiple coefficient + (income coefficient) × (slope of log income on police, same countries). Does it reproduce the simple coefficient? Explain the sign of the difference.

3. Repeat Task 1 with recorded burglaries. What happens to the police coefficient? How many countries are used, and why?
4. Build the four income groups. Compare the means with `tabstat`, then regress time on `i.income4` and on `ib4.income4`. Write the coefficient on High (first regression) and on Low (second) as sentences. Why are they equal with opposite sign?
5. Regress time on the quality of judicial processes index, then add `i.income4`. Does the coefficient on judicial quality change much? And the coefficient on High?
6. Write one main comment and one minor comment on Table 2, column (1), of Voigt and Wulf (2019).

Extension. Residualise both theft and police on log income and regress one residual on the other. Compare slope and standard error with the multiple regression of Task 1.

Self-check

- Did every coefficient of a multiple regression say what is held fixed?
- Did I read an indicator coefficient as a difference from the base group, not as a group mean?
- Did I avoid calling the coefficient with controls "the effect" of police?

Key definitions

- Multiple regression: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$. $\hat{\beta}_1$ is the difference in y associated with one more unit of x_1 among units with the same x_2.
- Partialling out: $\hat{\beta}_1$ equals the slope of y on the part of x_1 that x_2 does not predict (the residual of x_1 on x_2). Use the multiple regression for the standard error.
- Omitted variable bias: $\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \hat{\delta}_1$, where $\tilde{\beta}_1$ is the simple coefficient and $\hat{\delta}_1$ the slope of x_2 on x_1. The bias has the sign of $\hat{\beta}_2 \times \hat{\delta}_1$.
- Good control: determined before the variable of interest and related to the outcome. Bad control: itself affected by the variable of interest or produced by the same measurement process as the outcome.
- Indicators: with K groups, include $K - 1$ indicators and a constant. The constant is the mean of the base group; each coefficient is a difference from it. Changing the base group does not change the group means, the fitted values or R^2.
- Referee report: summary in your own words; numbered main comments (title, issue with its exact location, why it matters for a conclusion, what would resolve it); suggestions and minor comments in a separate, clearly labelled section; overall assessment.

Lab 6: assumptions, influential observations and changing samples

Question. When can we trust a regression's standard errors, and what can one country or a change of sample do to a coefficient?

Worksheet. `exercises/06_assumptions_and_samples.do`.

Before class: reading

Voigt and Wulf (2019), second reading: Section 8 in full; all columns of Table 2 and the footnotes. Bring notes on:

1. How many countries are used in each column of Table 2, and which variables explain the changes?
2. Which coefficients does Section 8 describe as robust across columns?
3. What does footnote 10 conclude from the Breusch–Pagan test?

Tasks

1. Regress theft on police and log GDP per capita. Draw residuals against fitted values, run `estat hettest`, and rerun with `vce(robust)`. Which numbers change and which do not? Does any conclusion change?
2. Compute Cook's distance and list the countries above $4/n$. Rerun without the most influential country. Which country is it, and why might it be unusual? Report both estimates in one sentence.
3. Add recorded burglaries as a second control, then rerun the first model with `if e(sample)`. How many countries does each regression use, and which leave? Split the change in the police coefficient into a sample part and a control part. Why are recorded burglaries a bad control?
4. Express the police coefficient in units, in standard deviations and as an elasticity at the means.
5. Regress the cost of enforcement on judicial quality in the 191 economies. Where is the spread of the residuals larger? By how much does the standard error change with `vce(robust)`? Does the conclusion change?
6. Rewrite the comment on Voigt and Wulf you wrote in class with the four steps, and write one comment of the same kind on the paper you are reviewing.

Self-check

- Did I avoid reading a non-significant heteroskedasticity test as proof of constant variance?
- Did I report the estimate with all observations, and name the observations I removed in a check?
- Did I compare two models on the same sample before attributing a change to a control?

Key definitions

- The six assumptions: (1) linear in the coefficients; (2) independent observations; (3) no perfect collinearity; (4) zero conditional mean of u_i; (5) constant variance; (6) normal errors.

Assumption 4 concerns the coefficients and their causal reading; assumptions 2 and 5 concern the standard errors.

- Heteroskedasticity: the spread of the residuals changes with x. Coefficients are unaffected; the usual standard errors are wrong. Robust standard errors (`vce(robust)`) do not require constant variance and do not fix dependence between observations or a biased coefficient.
- Outlier: a large residual. Leverage: an unusual value of x. Influence: removing the observation changes the coefficient a lot. Cook's distance summarises influence; a common flag is $D_i > 4/n$.
- Changing samples: every variable with missing values removes rows. To isolate the effect of a control, estimate both models on the same sample (`if e(sample)`) and report N in every column.
- Magnitude: in units; in standard deviations ($\hat{\beta} \times s_x/s_y$); as an elasticity at the means ($\hat{\beta} \times \bar{x}/\bar{y}$). Significance says whether a coefficient is distinguishable from zero, not whether it is large.
- A referee comment in four steps: where (table, column, page, footnote); what (a fact you can check); why it matters (which conclusion depends on it); what would resolve it (a feasible check, more information, or a narrower claim).